
PRACTICE FINAL EXAMINATION

MAT 1013 – DISCRETE MATHEMATICS

Instructions

1. Answer all four questions. Each main question worth 25 marks for a total of 100 marks.
 2. Questions 1 and 2 each contain 10 multiple-choice questions. Select the best answer in each case.
 3. **For Question 3, reproduce the given structure in a separate answer booklet and complete all blanks, matrices, tables, and intermediate steps as indicated.**
 4. For Question 4, present complete arguments and all necessary working.
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Question 1: Multiple-Choice Questions

1. The negation of

$$(p \wedge q) \rightarrow r$$

is logically equivalent to

- | | |
|------------------------------------|------------------------------------|
| a) $(\neg p \vee \neg q) \wedge r$ | d) $\neg p \wedge \neg q \wedge r$ |
| b) $p \wedge q \wedge \neg r$ | e) $p \vee q \vee \neg r$ |
| c) $p \wedge (q \rightarrow r)$ | |

2. Which of the following are tautologies?

- | | |
|--------------------------------|----------------------------|
| I. $\neg(p \wedge \neg p)$ | III. $\neg(p \vee \neg p)$ |
| II. $p \leftrightarrow \neg p$ | IV. $p \rightarrow p$ |
| a) I and II only | d) II and IV only |
| b) I and III only | e) I, II and IV only |
| c) I and IV only | |

3. The negation of

$$\exists x \in \mathbb{R} \forall y \in \mathbb{R}, \quad x \leq y$$

is logically equivalent to

- | | |
|---|--|
| a) $\exists x \in \mathbb{R} \exists y \in \mathbb{R}, x > y$ | d) $\exists y \in \mathbb{R} \forall x \in \mathbb{R}, x > y$ |
| b) $\forall x \in \mathbb{R} \exists y \in \mathbb{R}, x > y$ | e) $\forall y \in \mathbb{R} \exists x \in \mathbb{R}, x \leq y$ |
| c) $\forall x \in \mathbb{R} \forall y \in \mathbb{R}, x > y$ | |

4. The greatest common divisor of 252 and 198 is

- | | |
|-------|-------|
| a) 6 | d) 27 |
| b) 9 | e) 54 |
| c) 18 | |

5. Which of the following arguments are valid?

I. $p \rightarrow q, \neg q ; \therefore \neg p$

III. $p \vee q, \neg p ; \therefore q$

II. $p \rightarrow q, q ; \therefore p$

IV. $p \rightarrow q, \neg p ; \therefore \neg q$

a) I only

d) II and IV only

b) III only

e) All four

c) I and III only

6. Which of the following is congruent to 37 modulo 23?

a) $2^{35} + 3^{20}$

d) $2^8 + 5^{31}$

b) $5^{18} - 2^{11}$

e) $11^{12} + 6^7$

c) $7^{14} + 4^9$

7. The least non-negative solution of $9x \equiv 12 \pmod{26}$ is:

a) 3

d) 14

b) 8

e) 22

c) 10

8. The solution set of $|3x - 2| \geq 7$ is:

a) $[-\frac{5}{3}, 3]$

d) $(-\infty, -\frac{5}{3}] \cup [3, \infty)$

b) $(-\frac{5}{3}, 3)$

e) $(-\infty, \frac{5}{3}] \cup [3, \infty)$

c) $(-\infty, -3] \cup [\frac{5}{3}, \infty)$

9. Which of the following statements is/are **true**?

I. Every irrational number is a real number.

II. The sum of two irrational numbers is always irrational.

III. The product of a nonzero rational number and an irrational number is irrational.

IV. Every non-terminating decimal is irrational.

a) I and II only

d) I, III and IV only

b) I and III only

e) All four statements

c) II and IV only

10. The value of

$$\left(\frac{1 + \sqrt{3}i}{1 - i} \right)^{15}$$

is:

a) $128 + 128i$

d) $128 - 128i$

b) $-128 + 128i$

e) -256

c) $-128 - 128i$

Question 2: Multiple-Choice Questions

1. If $|A| = 3$ and $|B| = 5$, then the number of elements in $\mathcal{P}(A \times B)$ is

- a) 2^8
- b) 2^{15}
- c) 15^2
- d) $2^3 + 2^5$
- e) 3^5

2. Which identity is valid for all subsets A and B of a universal set U ?

- a) $(A \cup B)^c = A^c \cup B^c$
- b) $(A \cap B)^c = A^c \cap B^c$
- c) $(A \cup B)^c = A^c \cap B^c$
- d) $A \setminus B = B \setminus A$
- e) $A \cup (A \cap B) = B$

Items 3 and 4 are based on the relation

$$R = \{(1, 2), (2, 1), (2, 3), (3, 3)\}$$

on $A = \{1, 2, 3\}$.

3. The matrix of R , using the order 1, 2, 3 for rows and columns, is

- a) $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
- b) $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$
- c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$
- d) $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$
- e) None of the given

4. The inverse relation R^{-1} is

- a) $\{(1, 2), (2, 1), (2, 3), (3, 3)\}$
- b) $\{(2, 1), (1, 2), (3, 2), (3, 3)\}$
- c) $\{(2, 1), (1, 2), (2, 3), (3, 3)\}$
- d) $\{(1, 1), (2, 2), (3, 3)\}$
- e) None of the given

5. Let D be the divisibility relation on $\{1, 2, 4, 8\}$. Which statement is correct?

- a) D is an equivalence relation.
- b) D is reflexive, antisymmetric, and transitive, but not symmetric.
- c) D is symmetric and transitive, but not reflexive.
- d) D is reflexive and symmetric, but not transitive.
- e) D is irreflexive and antisymmetric.

6. Let

$$S = \{(1, a), (2, b), (3, a)\} \subseteq \{1, 2, 3\} \times \{a, b\}$$

and

$$T = \{(a, x), (a, y), (b, y)\} \subseteq \{a, b\} \times \{x, y\}.$$

Then $T \circ S$ is

- a) $\{(1, x), (2, y), (3, x)\}$
- b) $\{(1, x), (1, y), (2, y), (3, x), (3, y)\}$
- c) $\{(a, 1), (b, 2), (a, 3)\}$
- d) $\{(1, a), (2, b), (3, a), (a, x), (a, y), (b, y)\}$
- e) None of the given

The next four questions are based on the following relations on $A = \{1, 2, 3\}$:

$$R = \{(1, 1), (2, 2), (3, 3)\},$$

$$S = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3)\},$$

$$T = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}.$$

7. Which of the above relations are **reflexive** on A ?

- a) R and S only
- b) R and T only
- c) S and T only
- d) R only
- e) All three

8. Which of the above relations are **symmetric** on A ?

- a) R and S only
- b) R and T only
- c) S and T only
- d) R only
- e) All three

9. Which of the above relations are **transitive** on A ?

- a) R and S only
- b) R and T only
- c) S and T only
- d) R only
- e) All three

10. Which of the following is the composition $T \circ S$?

- a) $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3)\}$
- b) $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$
- c) $\{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3)\}$
- d) $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
- e) None of the given

Question 3: Structured Question

Part 1:

(a) Apply the Euclidean algorithm to 157 and 36 by completing the blanks:

$$\begin{aligned}
 157 &= 36(4) + \underline{\hspace{2cm}}, \\
 36 &= \underline{\hspace{2cm}}(2) + 10, \\
 13 &= 10(1) + \underline{\hspace{2cm}}, \\
 10 &= 3(\underline{\hspace{2cm}}) + 1, \\
 3 &= 1(\underline{\hspace{2cm}}) + 0.
 \end{aligned}$$

(b) Hence state $\gcd(157, 36)$.

(c) Work backwards through the Euclidean algorithm by completing the blanks to express $\gcd(157, 36)$ in the form $157x + 36y$:

$$\begin{aligned}
 1 &= 10 - 3(\underline{\hspace{2cm}}) \\
 &= 10 - 3(13 - \underline{\hspace{2cm}}) \\
 &= \underline{\hspace{2cm}}(10) - 3(13) \\
 &= 4(36 - 2(13)) - 3(13) \\
 &= 4(36) - \underline{\hspace{2cm}}(13) \\
 &= 4(36) - 11(157 - 4(36)) \\
 &= \underline{\hspace{2cm}}(36) - 11(157).
 \end{aligned}$$

(d) Use the identity obtained in part (c) to find the multiplicative inverse of 36 modulo 157.

From part (c),	$1 = \underline{\hspace{2cm}}(36) - \underline{\hspace{2cm}}(157).$
Therefore,	$\underline{\hspace{2cm}}(36) \equiv 1 \pmod{157}.$
Hence,	$36^{-1} \equiv \underline{\hspace{2cm}} \pmod{157}.$

(d) Complete the solution of

$$36u \equiv 19 \pmod{157}.$$

Using the multiplicative inverse obtained in part (c),

$$u \equiv \underline{\hspace{1cm}} \cdot 19 \equiv \underline{\hspace{1cm}} \pmod{157}.$$

Thus,

$$u = \underline{\hspace{1cm}} + 157k, \quad k \in \mathbb{Z},$$

and the least non-negative solution is

$$u = \underline{\hspace{1cm}}.$$

Part 2:

(a) Let

$$z_0 = -4 + 4\sqrt{3}i.$$

Complete the polar representation:

$$|z_0| = \underline{\hspace{1cm}}, \quad \text{Arg}(z_0) = \underline{\hspace{1cm}},$$

and hence

$$z_0 = \underline{\hspace{1cm}} (\cos \underline{\hspace{1cm}} + i \sin \underline{\hspace{1cm}}).$$

(b) Using De Moivre's theorem, calculate z_0^3 by completing the following steps:

From part (a),

$$z_0 = 8 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right).$$

Therefore,

$$\begin{aligned} z_0^3 &= (\underline{\hspace{1cm}})^{\square} (\cos(\underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}) + i \sin(\underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}})) \\ &= \underline{\hspace{1cm}} (\cos \underline{\hspace{1cm}} + i \sin \underline{\hspace{1cm}}) \\ &= \underline{\hspace{1cm}}. \end{aligned}$$

(c) Determine all three solutions of

$$z^3 = z_0$$

in polar form by completing the following steps:

Let

$$z = r(\cos \theta + i \sin \theta).$$

Then, by De Moivre's theorem,

$$z^3 = r^3(\cos 3\theta + i \sin 3\theta).$$

Since

$$z_0 = \text{_____} (\cos \text{_____} + i \sin \text{_____}),$$

we obtain

$$r^3 = \text{_____}, \quad 3\theta = \text{_____} + 2k\pi.$$

Hence,

$$r = \text{_____}, \quad \theta = \text{_____} + \text{_____}, \quad k = \text{_____}, \text{_____}, \text{_____}.$$

Complete the following table of solutions:

k	θ	z
_____	_____	_____
_____	_____	_____
_____	_____	_____

(d) Plot the solutions in the complex plane:

Part 3:

Let

$$A = \{1, 2, \dots, 6\}$$

and let R be the congruence modulo 4 relation on A :

$$aRb \iff a \equiv b \pmod{4}.$$

(a) With the elements ordered as 1, 2, 3, 4, 5, 6, complete the relation matrix:

$$M_R = \begin{pmatrix} \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \end{pmatrix}.$$

(b) Write **True** or **False** in each blank:

Property	True/False
R is reflexive	_____
R is symmetric	_____
R is antisymmetric	_____
R is transitive	_____

Part 4:

Define $f : \mathbb{Q} \setminus \{0\} \rightarrow \mathbb{Q}$ by

$$f(a) = \frac{18}{a}.$$

(a) Classify the function by completing the following table.

Property	True/False
R is injective	_____
R is surjective	_____
R is bijective	_____

(b) Compute $(f \circ f)(a)$.

$$(f \circ f)(a) = \underline{\hspace{2cm}}.$$

Question 4: Essay-Type Questions

(a) Solve the equation $z^5 = i$ and plot the answer on the complex plane.

(b) Consider the following relations:

- R given by $aRb \iff a|b$ for $a, b \in \{1, 2, \dots, 5, 6\}$.
- S be the relation from $\{1, 2, \dots, 5, 6\}$ to $\{T, F\}$, where $aST \iff a$ is even and $aSF \iff a$ is odd.

1. Draw the directed graph of R .
2. Verify that R is reflexive, anty-symmetric and transitive.
3. Write down the relaion matrices for the relations R and S .
4. What can you say about $R \circ R^{-1}$. Justify your answer.

5. Determine the matrix of $S \circ R$ using the matrices of R and S .

Show your work.

6. Write the domain, codomain and range of the relation $S \circ R$.

7. Determine $(S \circ R)^{-1}(T)$.

(c) Using a suitable method, prove:

For every integer n , if $3 \mid n^2$, then $3 \mid n$.

Clearly state results used in the proof and justify your work .

(d) Prove using mathematical induction that

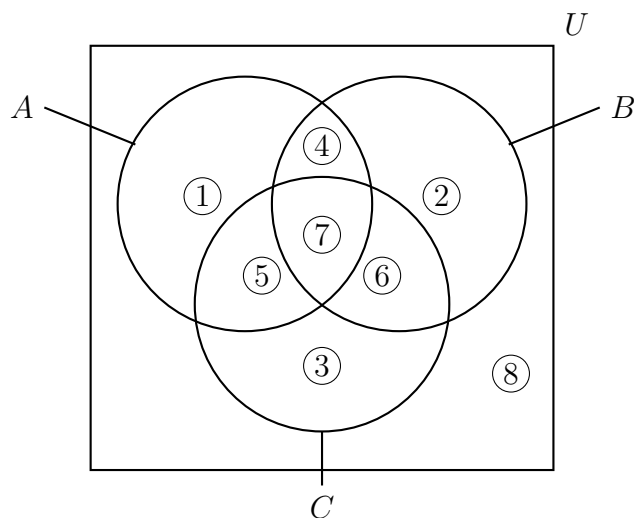
$$\sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}$$

for every positive integer n .

(e) Using the labelled Venn diagram below, verify

$$(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C).$$

Important: The numbered labels represent regions, not elements. Do not use set braces around the region labels.



MAT 1013 Discrete Mathematics
MCQ Answer Sheet for Questions 1. and 2.

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